

Paper 5 -> Learning Analytics Dashboard Applications (Verbert et al. 2013)

Learning analytics dashboards visualize learning traces (data about student activity) to help teachers and learners become aware of learning patterns, reflect on progress, make sense of data, and ultimately change behavior or improve learning. The paper presents a conceptual framework for analyzing dashboard applications and reviews 15 existing systems.

The Four-Stage Process Model:

1. **Awareness** — Just presenting the data (activity streams, tables, visualizations). What did students do?
2. **Reflection** — Users ask questions about the data. Is this useful? What does it mean?
3. **Sensemaking** — Users answer their own questions and create insights. Why did this happen?
4. **Impact** — Change behavior or meaning based on insights. What will I do differently?

What Gets Tracked in Dashboards:

- **Time spent** — 9 of 15 systems track this
- **Social interaction** — 8 systems, mainly to detect isolated/collaborative students
- **Document and tool use** — 10 systems, to measure effort and find popular resources
- **Produced artifacts** — 10 systems (blog posts, forum posts, etc.)
- **Assessments** — 9 systems track exercise/quiz/assessment performance

Examples of Existing Dashboards:

- **Course Signals** (Purdue) — only system with proven impact; students using dashboard had 96.71% retention vs 83.44% without
- **CALMSystem** — helps learners reflect and improve self-assessment
- **GLASS** — tracks collaborative learning
- **Student Activity Meter (SAM)** — tracks effort indicators
- **StepUp!** — visualizes learner activity and contributions for both teachers and students

Key Finding on Effectiveness:

Only **Course Signals** demonstrated actual behavior change (Stage 4 impact). Most other evaluations only measure:

- Perceived usefulness (questionnaires)
- Stage 1-3 (awareness, reflection, sensemaking)
- Whether teachers could identify at-risk students

Critical Limitation:

Most dashboards fail to measure whether they actually **change behavior or improve learning**. This requires longitudinal studies. Teachers perceived higher usefulness than students. Students perceived higher usefulness when dashboards showed **complete** activity data (e.g., time + social interaction together better than just one metric).

Dashboard Design Principles:

- Visualize learner effort through multiple data types
- Track both individual and collaborative activity
- Show progress over time (current week vs. overall)
- Present data in multiple views (teacher view vs. learner view)
- Use "modest computing" — empower users to make decisions, don't automate decisions for them

The "Quantified Self" Connection:

Dashboards work best when tied to goals and progress toward those goals. Meaningful feedback loops sustain desired behavior this is the core of the quantified self movement in learning.